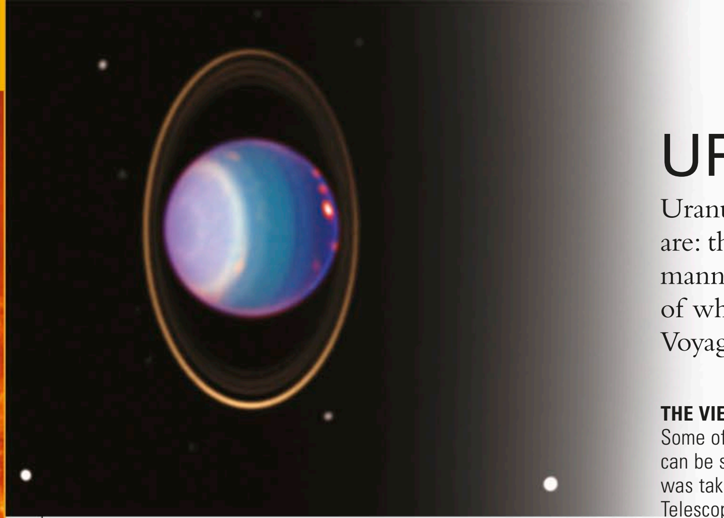


URANUS'S MOONS

Uranus's moons can be divided into three groups. Moving out from Uranus, they are: the small inner satellites; the five major moons, which orbit in a regular manner; and the small outer moons, many of which follow retrograde orbits. Much of what is known about the moons, and the only close-up views, came from the Voyager 2 flyby in 1985–1986. This revealed the major moons to be dark, dense rocky bodies with icy surfaces, featuring impact craters, fractures, and volcanic water-ice flows. The moons are named after characters in the plays of the English dramatist William Shakespeare or in the verse of the English poet Alexander Pope.

THE VIEW FROM EARTH
Some of the 27 moons that orbit Uranus can be seen in this infrared image, which was taken by the Hubble Space Telescope in 1998.

INNER MOON

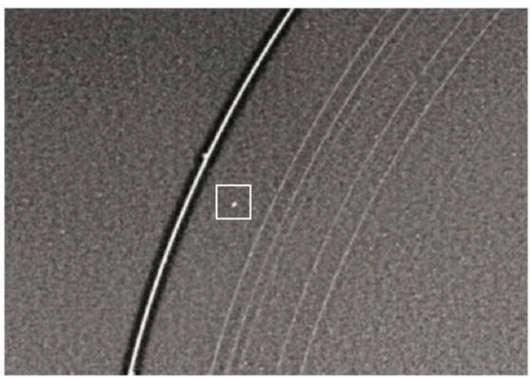
Cordelia

DISTANCE FROM URANUS 30,910 miles (49,770 km)

ORBITAL PERIOD 0.34 Earth days

DIAMETER 25 miles (40 km)

Cordelia is the innermost and one of the smallest of Uranus's moons. A team of Voyager 2 astronomers discovered it on January 20, 1986. Cordelia was one of 10 moons that were discovered in the weeks between December 30, 1985, and January 23, 1986, as the Voyager 2 spacecraft flew by Uranus and transmitted images back to Earth. Astronomers had expected to find some more moons in orbit around Uranus. In particular, it was expected that pairs of shepherd moons—moons that are positioned on either side of a ring and keep the ring's constituent particles in place—would be found. Surprisingly, just one pair, that of Cordelia and Ophelia, was discovered. Cordelia takes its name from the daughter of Lear in Shakespeare's *King Lear.*



SHEPHERD MOON
Cordelia is the innermost of two shepherd moons lying on either side of Uranus's bright outer ring.

INNER MOON

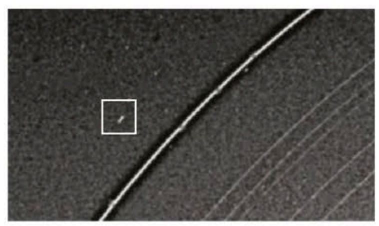
Ophelia

DISTANCE FROM URANUS 33,400 miles (53,790 km)

ORBITAL PERIOD 0.38 Earth days

DIAMETER 26 miles (42 km)

Ophelia is one of a pair of moons that orbit either side of Uranus's outer ring, the epsilon ring. It was discovered at the same time as its partner, Cordelia, on January 20, 1986. The two are small—not much bigger than the particles that make up the thin, narrow ring. The moon is named after the heroine in Shakespeare's *Hamlet.*



OPHELIA LIES OUTSIDE THE EPSILON RING

INNER MOON

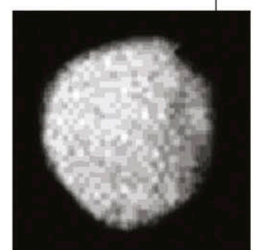
Puck

DISTANCE FROM URANUS 53,410 miles (86,010 km)

ORBITAL PERIOD 0.76 Earth days

DIAMETER 101 miles (162 km)

Puck was discovered on December 30, 1985, and was the first of the 10 small moons to be found in the Voyager 2 data. It is the second-farthest inner moon from Uranus and was discovered as the probe approached the planet. There was time to calculate that an image could be recorded on January 24, the day of closest approach. The image (above) revealed an almost circular moon with craters. The largest crater (upper right) is named Lob, after a British Puck-like sprite.



CRATERED MOON

MAJOR MOON

Miranda

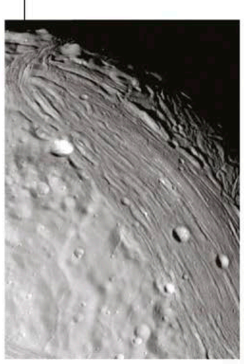
DISTANCE FROM URANUS 80,350 miles (129,390 km)

ORBITAL PERIOD 1.41 Earth days

DIAMETER 300 miles (480 km)

Miranda is the smallest and innermost of Uranus's five major moons and was discovered by Dutch-born American astronomer Gerard Kuiper on February 16, 1948. When all five major moons were seen in close-up for the first time, on January 24, 1986, it was Miranda that gave astronomers the biggest surprise. As Voyager 2 passed within 19,870 miles (32,000 km) of its surface, the probe revealed a bizarre-looking world where various surface features butt up against one another in a seemingly unnatural way. One explanation for this strange appearance is that Miranda experienced a catastrophic collision in its past. The moon shattered into pieces and then reassembled in the disjointed way seen today. An alternative theory says that the moon's evolution was halted before it could be completed. Soon after its formation, dense, rocky material began to sink and lighter material, such as water ice, rose to the surface. This process then stopped because the necessary internal heat had disappeared. The surface clearly has different types of terrain from different time periods.

FULL DISK
The complex terrain of the bright, chevron-shaped Inverness Corona stands out in this south polar view of Miranda.



GEOLOGICAL MIX
On the left lies an ancient terrain of rolling hills and degraded craters; to the right is a younger terrain of valleys and ridges.

CRATERS AND FAULTS
Many different-sized impact craters can be seen in this 125-mile (200-km) wide region of rugged, high-elevation terrain, indicating that it is older than the lower terrain. Faults cut across the terrain at lower right.

THE SOLAR SYSTEM